MANGROVE NON TIMBER PRODUCT

Guideline for ASEAN Strategy on Mangrove Ecosystem Rehabilitation

Series 7

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FOREWORD

Mangrove forests are among the world's most valuable coastal ecosystems, providing an extensive range of ecological, economic, and cultural benefits to millions of people in the ASEAN region. Beyond their role in shoreline protection, biodiversity conservation, and carbon sequestration, mangroves offer a rich variety of non-timber products that have sustained coastal communities for generations. These include food ingredients, traditional medicines, natural dyes, handicrafts, and other goods that hold significant potential for local enterprise development when managed responsibly.

The sustainable utilization of mangrove non-timber products represents an opportunity to strengthen community livelihoods while safeguarding ecosystem health. By promoting eco-friendly harvesting, value-added processing, and equitable benefit sharing, these resources can become catalysts for economic resilience and environmental stewardship. This aligns closely with the ASEAN Strategy on Mangrove Ecosystem Management, particularly Strategy No. 2: Promoting the urgency, managing knowledge, and enhancing capacity for conservation, protection, restoration, and sustainable use of mangrove ecosystems in the ASEAN region.

This technical guide has been developed to serve as a practical reference for communities, practitioners, local governments, and development partners. It provides methods, tools, and best practices for the processing and utilization of mangrove-based materials, underpinned by a strong commitment to conservation and sustainable management.

We hope this guide will inspire collective action, strengthen local capacity, and contribute to a future where mangrove ecosystems are both protected and productively integrated into the sustainable development of ASEAN's coastal regions.

Author

Indonesia, 2025

I.

INTRODUCTION

Mangrove forests are among the most productive and biologically diverse coastal ecosystems. In addition to their critical role in protecting shorelines from erosion and preventing seawater intrusion, mangroves also serve as highly effective carbon sinks. These ecosystems provide natural habitats for more than 1,533 species of fish and marine organisms, functioning as vital nursery grounds for various commercially valuable fish species. Mangrove forests also support the health of surrounding ecosystems, such as coral reefs and seagrass beds (UNEP, 2023).

Beyond their ecological value, mangrove forests hold significant economic potential, particularly through the sustainable use of **non-timber forest products (NTFP)**. NTFP from mangroves refer to various goods derived from mangrove ecosystems—excluding timber—that hold economic value and are commonly utilized by local communities. These include fruits, leaves, bark, and honey produced by bees living in and around mangrove areas. Such products can be processed into value-added goods such as syrup, jam, herbal tea, natural dyes, and traditional medicines (BPHM, 2007). The utilization of these NTFP offers opportunities for small-scale enterprises and alternative income sources for coastal communities, without the need to cut down or damage mangrove forests.

However, the utilization of NTFP from mangrove ecosystems must be carried out sustainably to ensure the continued ecological function of these forests for both current and future generations.

This requires sufficient knowledge and appropriate technical skills throughout the harvesting, processing, and marketing stages, so that the exploitation of mangrove resources aligns with environmental conservation principles.

This guidebook has been prepared as a practical reference for coastal communities, small business groups, and environmental practitioners interested in developing mangrove-based NTFP enterprises. It contains comprehensive information on the types of NTFP available, environmentally friendly harvesting methods, simple processing techniques, and product marketing strategies. forests, which are essential to the sustainability of coastal life.

II.

MANGROVE ECOSYSTEM

A. Definition of Mangrove

Mangroves are plant species that grow in tropical and subtropical coastal areas, particularly in tidal zones (FAO, 1982). According to the Decree of the Minister of Environment (2004), mangroves are a group of plants consisting of various species with taxonomic relationships up to the class level, but which share morphological and physiological adaptations to tidal-influenced habitats. This ecosystem thrives in intertidal areas dominated by woody plants that can survive in muddy soils with high salinity and are periodically inundated by seawater (Djamaluddin, 2018).

Mangroves play a crucial role in protecting coastlines, filtering pollutants, absorbing carbon, and providing habitat for various species of fish, shrimp, birds, and other marine life.

B. Ecological Functions of Mangroves

Mangrove forests have vital ecological functions for maintaining the balance of coastal environments. As a transitional ecosystem between land and sea, mangroves play a key role in ensuring ecosystem stability and supporting life both on land and in the ocean. The following are some of the ecological functions of mangroves:

1. Coastal Protection
   - » The sturdy and dense roots of mangroves act as natural barriers against ocean waves, reducing the risk of coastal erosion and damage from storms or tsunamis.
   - » Mangrove roots trap soil, preventing erosion and helping to stabilize shorelines.
2. Habitat and Nursery Grounds
   - » Mangroves provide habitats, breeding grounds, and shelter for various species of fish, crabs, shrimp, shellfish, and birds.
   - » Many traditional fishers rely on the presence of mangrove forests for their catch.
3. Carbon Sequestration and Oxygen Production
   - » Mangroves are highly effective in absorbing carbon dioxide (CO_2) from the atmosphere—even more so than terrestrial tropical forests.
   - » They help mitigate the impacts of climate change through carbon sequestration.
4. Natural Water Filtration
   - » Mangrove roots help filter incoming water by trapping sediments and pollutants, thereby maintaining seawater quality in the surrounding area.
5. Sustainable Natural Resources
   - » Mangroves produce non-timber forest products (NTFP) such as fruits, leaves, honey, and raw materials for natural medicines, without the need to destroy the trees.

C. Common Types of Mangroves in Indonesia

Indonesia's mangrove biodiversity ranks among the richest in the world, both in terms of area and number of species. There are over 200 mangrove species recorded in the country, with about

47 classified as true mangroves, while the rest are associated mangroves—plants that grow in coastal areas but are not entirely dependent on tidal conditions. Below are four common types of mangroves in Indonesia that have high ecological and economic value:

1. *Rhizophora mucronata* (Bakau)

Characteristics:

- Prominent stilt roots extending from the trunk.
- Broad leaves, dark glossy green on top, and pale green underneath.
- Produces long pencil-shaped propagules that hang before falling into the mud and growing into new seedlings.

Habitat:

- Grows in the middle to front zones of tidal areas, in deep and stable mud.

Benefits:

- Bark is rich in tannins, used in leather tanning or as a natural dye.
- Roots and leaves serve as habitats for crabs and small fish.

2. *Avicennia marina* (Api-api)

Characteristics:

- Small leaves, silvery green, with fine hairs on the underside.
- Pneumatophores (aerial roots) rise from the mud like short sticks.

Habitat:

- Typically found at the outermost edge of the mangrove zone, near open sea.

Benefits:

- Leaves are used in traditional medicine (anti-inflammatory, antiseptic).
- Leaf powder can be used as organic fertilizer.
- Roots help stabilize fine sediment and provide habitat for plankton.

3. *Sonneratia alba* (Pidada/Bogem)

Characteristics:

- Pneumatophores like small tubes surrounding the base of the trunk.
- Leaves are oval and slightly thick.
- White flowers with long stamens; produces large round fruits.

Habitat:

- Grows along the coastline on muddy or sandy soils.

Benefits:

- Its fruit, known as pidada or bogem, can be consumed or processed into syrup, jam, and traditional sweets (dodol)
- Flowers are a nectar source for bees that produce mangrove honey.

4. *Bruguiera gymnorrhiza* (Lindur)

Characteristics:

- Has knee roots that arch above the mud surface.
- Oval leaves and large, striking red flowers.
- Produces elongated fruits with pointed tips, resembling small spears.

Habitat:

- Commonly found in the middle zone of mangroves alongside Rhizophora species.

Benefits:

- Bark contains high levels of tannins, used as natural dye and preservative.
- Leaves and stems are used in traditional medicine.
- Suitable as natural dye for batik and as an ingredient in herbal cosmetics.

III.

TYPES OF NON-TIMBER FOREST PRODUCTS (NTFP) FROM MANGROVES

Mangrove ecosystems not only provide critical ecological functions but also offer substantial economic potential through the utilization of non-timber forest products (NTFP). These NTFP include a variety of natural resources that can be harvested without cutting down trees, thus supporting both livelihoods and ecosystem sustainability. Utilizing mangrove NTFP presents an opportunity for coastal communities to diversify their income sources while actively contributing to mangrove conservation.

With adequate understanding of the types of NTFP, their characteristics, and processing potentials, communities can develop environmentally friendly businesses based on mangrove resources. This section explores some common mangrove-derived NTFP in Indonesia that offer promising opportunities for development.

A. Mangrove Fruits (Pidada and Nypa Palm)

1. *Sonneratia caseolaris* (Pidada/Bogem)

Characteristics:

- The fruit is round, green in color, with a distinctive sour flavor.
- Commonly grows on trees located at the edge of the mangrove zone near open waters.
- Utilized by both wildlife and local communities.

Figure 1. Fruit of *Sonneratia caseolaris* and Its Processed Products (Syrup)

Utilization Potential:

- The fruit can be processed into various value-added products such as syrup, jam, traditional sweets (dodol), and candied fruit.
- Rich in antioxidants and vitamins, making it a suitable ingredient for health foods and herbal beverages.

Utilization:

- Raw material for food and beverage products such as syrup, jam, and traditional sweets (dodol).
- Rich in vitamin C and antioxidants.

Product Examples:

- Pidada/Bogem Syrup, Pidada Jelly Candy, and Mangrove Dodol.

2. *Nypa Fruticans* (Buah Nipa palm)

Characteristics:

Grows in clusters at river estuaries; the fruit contains soft and sweet seeds.

Figure 2. Fruit of *Nypa fruticans* and Its Processed Products (Sugar)

Utilization:

- The soft, sweet seeds are commonly used as ingredients in refreshing drinks such as mixed ice desserts and are also processed into nipa palm dodol (traditional sweets) and nipa palm sugar.
- Sap extracted from the flower stalks can be fermented or cooked into high-value products such as nipa palm vinegar, nipa palm sugar (in block or liquid form), and natural crystal sugar (palm sugar granules).

Product Examples:

Nipa palm fruit iced dessert, nipa palm sugar blocks, nipa palm liquid sugar, nipa palm vinegar, and nipa palm dodol.

B. Mangrove Forest Honey

Origin:

Produced by wild bees such as *Apis dorsata* (giant honeybee) or Trigona (stingless bee), which collect nectar from mangrove flowers—particularly from Sonneratia (pidada) and Avicennia (api-api) species.

Mangroves provide a rich and diverse nectar source, giving the honey a unique aroma and flavor distinct from regular floral honey.

Utilization:

- Known for its health benefits, including boosting immunity, anti-inflammatory properties, and serving as a natural energy source.
- It can be consumed directly, mixed into herbal drinks, or used as an ingredient in health and beauty products.

Product Examples:

Pure mangrove honey, herbal honey beverages, natural honey soap, and honey-based face masks.

Figure 3. Honey from Mangrove Forests

Characteristics: Sweet taste with a slight tang and a distinctive aroma of mangrove flowers.

Benefits: Natural energy sources contain antibacterial and antioxidant compounds.

Business Potential: Beekeeping (apiculture) in mangrove areas offers great potential and is environmentally friendly.

C. Natural Dyes and Tannins

Origin:

The bark of *Rhizophora* spp. and *Bruguiera spp*. trees contain high levels of tannins. However, harvesting bark is not recommended as it can lead to the exploitation and degradation of mangrove trees.

Alternatively, dried waste from mangrove fruits (*Rhizophora spp*.) can be utilized as a more sustainable source of natural dyes derived from mangroves.

Figure 4. Batik Dyed with Mangrove Propagule Extract (Batik Bakau)

Benefits:

- Tannins are used for leather tanning, preserving organic materials, and as a traditional remedy for wounds.
- The extract is used as a natural dye for batik fabrics, producing a brown color.

Product Examples:

Mangrove batik dye, herbal antiseptic, natural fish preservative.

D. Mangrove Leaves for Traditional Medicine

Type: *Acanthus* sp. Leaves

Figure 5. Processing Jeruju Leaves into The Herbal drink.

Utilization:

- Processed into herbal tea, diarrhea medicine, natural antiseptic, to ointment for external wounds. Contains antibacterial substances and bioactive compounds.

Product Example:

Mangrove leaf tea

IV.

SUSTAINABLE HARVESTING TECHNIQUES FOR NON-TIMBER FOREST PRODUCTS (NTFP) FROM MANGROVE ECOSYSTEMS

The utilization of Non-Timber Forest Products (NTFP) from mangrove forests must be conducted with care, wisdom, and a strong commitment to sustainability. This means that harvesting practices must ensure the survival of the trees, the preservation of the environment, and the balance of the ecosystem. The act of harvesting should never come at the cost of damaging or killing mangrove trees, which are invaluable to coastal ecosystems. Below are technical guidelines for the sustainable and environmentally friendly harvesting of mangrove-based NTFP:

A. Harvesting Mangrove Fruits (e.g., Sonneratia or Pedada and Nypa Palms)

1. Best Practices:
    - Only harvest ripe fruits, indicated by a change in color (usually darker) and ease of detachment from the stem.
    - Use picking tools or poles to collect fruits; avoid climbing or pulling branches roughly.
    - Collect only as much as needed or as can be processed based on available production capacity.

2. Practices to Avoid:
   - » Do not pick unripe fruits as they are not suitable for consumption and may disrupt the plant's reproductive cycle.
   - » Avoid damaging leaves, twigs, or trunks when harvesting fruits.
   - » Do not harvest all fruits from a single tree. Always leave a portion for natural regeneration and for wildlife food sources.

Note: A single Sonneratia tree can yield hundreds of fruits per season. With proper management, the yield is sufficient for household consumption or small-scale business use.

B. Harvesting Bark (for Natural Dyes and Tannins)

1. Best Practices:
   - » Collect bark from dead branches or trunks that have fallen naturally.
   - » If harvesting from live trees is absolutely necessary, remove only small sections and never encircle the trunk (to avoid girdling).
   - » Use sharp tools such as knives or machetes for clean cuts, minimizing damage to the tree's living tissue.
2. Practices to Avoid:
   - » Never strip bark completely from the main trunk of a living tree, as this can quickly kill the tree.
   - » Avoid cutting down trees solely for their bark.

Note: Species such as Rhizophora spp. are rich in tannins, which can be used for natural batik dye or leather tanning. However, harvesting must be done in controlled quantities to ensure forest sustainability.

C. Harvesting Mangrove Honey

1. Best Practices:
   - » Use artificial hives (stup) to cultivate honeybees around mangrove areas.
   - » When harvesting from wild colonies, apply light smoke to calm bees without destroying the hive.
   - » Harvest only mature honey and leave part of the hive intact to support colony survival.
2. Practices to Avoid:
   - » Do not burn or completely destroy bee nests.
   - » Avoid excessive harvesting from wild colonies, as this can force bees to abandon the site or cause colony collapse.
   - » Refrain from using chemicals that may harm bees or contaminate the environment.

Note: Honey produced by bees foraging in mangrove ecosystems has a distinctive aroma and high antioxidant content. Sustainable apiculture practices can yield high-quality honey while preserving mangrove health.

D. Harvesting Leaves and Flowers

1. Best Practices:
   - » Whenever possible, collect fallen or mature leaves and flowers.
   - » If harvesting from live trees, pick evenly from various branches and not from a single part.
   - » Use garden shears or pruners to ensure clean cuts and minimize plant injury.
2. Practices to Avoid:
   - » Do not remove too many leaves from one tree, as they are essential for photosynthesis.
   - » Do not harvest all flowers—leave enough for pollination and fruit production to maintain ecological balance.

Note: Leaves from Acanthus spp. are used in traditional medicine and herbal teas. Flowers of Sonneratia alba are important nectar sources for bees. Overharvesting can disrupt ecological relationships and reduce long-term benefits.

E. General Principles of Sustainable NTFP Harvesting

1. Harvest Modestly, Not Greedily. Focus on long-term gains rather than short-term exploitation.
2. Avoid Damaging the Tree. NTFP should be collected WITHOUT cutting down or killing the tree.
3. Use Proper Tools. Clean, sharp tools reduce excessive wounds on the plant and prevent infections.
4. Monitor Regrowth. Conduct regular observations to ensure that trees remain healthy and productive.
5. Involve the Community. Harvesting conducted by community groups or cooperatives is more manageable and accountable than unregulated individual harvesting.

V.

MARKETING OF NON-TIMBER FOREST PRODUCTS (NTFP) FROM MANGROVE ECOSYSTEMS

Non-timber forest products (NTFP) from mangrove ecosystems have significant economic potential. However, having a good product is not enough. For a business to grow and generate sustainable profit, a sound marketing strategy is essential. Marketing is not only about selling but also about building a product image, reaching consumers, and fostering customer loyalty.

A. What is Marketing?

Marketing is not merely an activity to sell products but a comprehensive strategic process. It begins by introducing the product to potential buyers through various communication channels, both conventional and digital. The process involves generating interest through relevant messaging, appealing visuals, and conveying the value of the product in line with the needs and desires of the target market. Once attention and interest are established, the goal is to persuade potential buyers to take action—namely, to purchase—by providing compelling rational and emotional reasons.

However, marketing does not stop at the first transaction. The next phase is about creating a positive customer experience through product quality, good service, and continuous communication. Customer satisfaction is the key to loyalty, eventually turning casual

buyers into repeat customers. Loyal customers not only continue to purchase but may also become advocates who promote the product to others.

In other words, marketing is a process of building a mutually beneficial long-term relationship between producers and consumers. Success is measured not only by how many products are sold but also by how strongly the brand is embedded in the minds and hearts of customers. Marketing includes promotion, pricing, product distribution, and consumer communication.

Examples of NTFP products with high market value:

NTFP Product	Potential Target Market
Pedada Fruit Syrup/Jam	Souvenir shops, local stores, tourists
Mangrove Honey	Herbal pharmacies, local honey enthusiasts
Natural Dyes	Batik artisans, small textile businesses
Mangrove Herbal Tea	Health market, organic communities
Nypa Vinegar	Healthy culinary market, restaurants
Nypa Leaf Crafts	Tourists, art markets, gift shops
Compost/Fermented Feed	Local farmers, agricultural cooperatives

B. Marketing Strategies for Mangrove NTFP Products

1. Know Your Product and Its Unique Selling Point (Positioning)
   - » Understand what makes your product special before selling.
   - » Examples:
   - » Natural Pedada syrup, preservative-free!
   - » Distinctive mangrove honey, high in antioxidants!
   - » Eco-friendly batik using mangrove-based natural dye!
   - » Tip: Create a short and catchy slogan that describes your product effectively.

2. Set an Appropriate Price (Pricing)
   a. Consider:
      ~ Production Costs (Materials, Labor, Packaging)
      ~ Profit Margin
      ~ Market Price
      ~ Consumer purchasing power
   b. Example: If production cost of mangrove tea per pack is Rp3,000, sell at Rp5,000–Rp7,000.
      ~ Tip: Offer wholesale discounts or introductory promotions.
3. Use Attractive and Informative Packaging
   a. Good packaging protects the product, grabs consumer attention, and conveys essential product info.
   b. Required label information:
      ~ Product name
      ~ Ingredients
      ~ Net weight
      ~ Production and expiration date
      ~ Manufacturer name and contact info
      ~ Halal or BPOM certification (if applicable)

   Tip: Use eco-friendly packaging to increase product value.
4. Choose Effective Marketing Channels (Place/Distribution)

   Distribution methods include:
   a. Direct Marketing:
      ~ Direct sales to consumers (markets, exhibitions, door-to-door, cooperatives)
      ~ Social media (WhatsApp, Facebook, Instagram)
      ~ Online marketplaces (Tokopedia, Shopee, etc.)

b. Indirect Marketing:
   - ~ Consignment at souvenir shops, stalls, cafes, or MSME centers
   - ~ Collaboration with cooperatives, village enterprises (BUMDes), or NGOs
   - ~ Supplying to hotels or local restaurants

   Tip: Create dedicated social media and e-commerce accounts for your product.

5. Promote Products Consistently (Promotion)
   - » Promotion methods and examples:
   - » Social Media: Upload product photos and customer testimonials
   - » Word of Mouth: Ask buyers to share with others
   - » Exhibitions: Join local bazaars, festivals, tourism events
   - » Discounts & Promos: Buy 3 get 1 free, Ramadan packages, etc.
   - » Product Education: Distribute brochures or leaflets about benefits and processes

   Tip: Encourage customers to leave reviews or testimonials to build trust.

C. Building a Product Brand for Mangrove NTFP

Branding is the identity that makes a product recognizable and memorable. Key elements of branding include:

- Product name (e.g., Coastal Honey, Lestari Mangrove Jam)
- Simple logo
- Signature packaging colors
- Unique backstory (storytelling)
- Tip: Share how the product is made by coastal communities that protect mangrove forests.

D. Collaboration and Marketing Networks

NTFP products grow faster when managed collectively (through groups or MSMEs). Possible networks include:

- Coastal women's cooperatives/groups
- Environmental or economic empowerment NGOs
- Village-owned enterprises (BUMDes)
- Local government offices (Cooperative, Trade, Tourism Departments)
- MSME marketplaces such as Bangga Buatan Indonesia

Tip: Look for partners with broader market access.

E. Evaluating and Improving Strategies

Marketing efforts should be evaluated continuously. Ask questions like:

- Which products sell the most?
- Where are the most effective sales locations?
- What is customer feedback?
- How do revenues compare to costs?
- Use simple records (e.g., monthly tables) to monitor progress.

VI.

CHALLENGES IN DEVELOPING NON-TIMBER FOREST PRODUCTS (NTFP) FROM MANGROVES

Mangrove-based Non-Timber Forest Products (NTFP) hold great potential to support coastal economies while maintaining ecological sustainability. However, the development of these products continues to face significant challenges. One key issue is limited knowledge among communities regarding the types and benefits of mangrove NTFP. Many are unaware that mangrove fruits, leaves, and even roots can be processed into high-value products such as natural dyes, herbal syrups, soaps, and cosmetics. The lack of innovation and minimal research efforts also hinder the full utilization of these resources.

Another major challenge is limited access to processing infrastructure and technology. Many local entrepreneurs do not have adequate tools or facilities to process mangrove NTFP hygienically and efficiently. Furthermore, as many mangrove areas are designated conservation zones, regulatory and permitting issues often arise. Local communities frequently struggle to gain legal access for sustainable resource use due to insufficient information or support.

a) Illustration: Mangrove planting activities as part of conservation efforts

The market for mangrove NTFP products is also relatively small and seasonal. These products often lack strong branding, making them less competitive in broader markets. Promotion and consumer education remain suboptimal. Simultaneously, ecological pressures such as land conversion into aquaculture or housing developments threaten the long-term availability of mangrove raw materials. This instability poses risks to NTFP-based businesses if not accompanied by robust ecosystem protection.

To address these challenges, a collaborative approach is essential. Communities, governments, academia, and private sectors must work together to provide training, enhance business capacities, and promote research and innovation. Developing green branding that highlights ecological and social values can increase market appeal. Legal access to mangrove resources can be facilitated through social forestry schemes, while market reach can be expanded via digital platforms and strategic partnerships.

With the right strategies, mangrove-based NTFP can provide not only economic value but also strengthen coastal ecosystem conservation. The development of products such as natural dye batik, herbal supplements, and functional foods from mangroves can serve as inclusive green entrepreneurship models. These initiatives open opportunities for local communities—especially women and youth—to play active roles in sustainable green value chains.

b) Illustration: Mangrove wood charcoal production as a form of mangrove exploitation

From a conservation standpoint, developing mangrove NTFP reflects the complex balance between economic use and ecosystem protection. On one hand, mangroves offer diverse high-value resources such as food, medicine, and natural dyes. On the other hand, they are highly sensitive ecosystems, crucial for coastal protection, carbon sequestration, and biodiversity.

Unsustainable harvesting practices—such as excessive leaf or fruit collection—without ecological guidelines can degrade mangrove health. This often occurs due to the absence of community-level technical guidelines for sustainable harvesting.

Climate change and pollution exacerbate these conservation challenges. Sea level rise, saltwater intrusion, and land-based pollutants (e.g., plastics and agricultural runoff) can impair mangrove growth, ultimately affecting NTFP productivity. Conservation efforts must become more adaptive, focusing not only on preserving vegetation but also maintaining supportive environmental conditions.

A further issue is the lack of integration between conservation and economic development programs. Many conservation initiatives are top-down and exclude local input, while economic projects tend to prioritize output over ecological limits. A solution lies in promoting ecosystem-based social forestry approaches to ensure a balanced, sustainable integration of conservation and NTFP utilization.

VII.

CASE STUDIES AND BUSINESS INSPIRATIONS

Mangrove NTFP management is more than theory—it's already being successfully practiced across Indonesia and beyond. Numerous coastal communities, cooperatives, and small entrepreneurs have developed NTFP-based enterprises grounded in conservation and local economic empowerment.

A. Objectives of Case Studies

1. Showcase successful examples of mangrove NTFP management.
2. Inspire other coastal communities to start their own ventures.
3. Demonstrate that conservation and income generation can go hand in hand.

1. Kampung Nipa palm, Serdang Bedagai – North Sumatra (Indonesia)

Flagship Products: Liquid Sugar and Vinegar from Nypa Sap

Actors: Mangrove Farmer Group – Kampung Nipa palm

Success Story:

Locals collect sap from naturally growing Nypa palms without felling the trees, and process it into liquid sugar, syrup, and vinegar. These eco-friendly, health-conscious products have become popular, and ecotourism activities supplement community income.

Keys to Success:

- Synergy of NTFP business and educational tourism
- Marketing with a conservation narrative
- Support from CSR programs and NGOs

2. Teluk Kelumpang, South Kalimantan (Indonesia)

Flagship Products: Mangrove Honey and Herbal Tea

Actors: Community Forest Group (HKm) – Teluk Kelumpang

Success Story:

Residents produce high-quality mangrove honey by cultivating local bees and process *Avicennia marina* leaves into packaged herbal tea. Their honey is sold in souvenir shops, and their tea is gaining traction in MSME markets.

Keys to Success:

- Integration of conservation, cultivation, and processing
- Professional packaging and lab testing
- Partnerships with forestry agencies and universities

3. Srikandi Pantura Batik Group – Semarang (Indonesia)

Flagship Products: Natural-Dye Batik from Mangrove Waste
Actors: Coastal Women Artisans in Mangunharjo

Success Story:

Using Rhizophora bark, they produce natural reddish-brown dyes for batik. The batik stands out as eco-friendly and chemical-free. It is now sold online at national expos, and even via limited exports.

Keys to Success:

- Eco-innovation in product development
- Strong storytelling and green branding
- Design training from creative communities.

4. Five Indigenous Tribes – Papua (Indonesia)

Flagship Products: Traditional Medicines from Mangrove Species
Actors: Mandender, Sough, Inanwatan, Ambay, and Mandori Tribes

Success Story:

These tribes use traditional knowledge to create herbal remedies from seven mangrove species, including *Bruguiera gymnorrhiza* and *Sonneratia alba*, treating ailments like scabies, malaria, and toothache. Research has validated their methods and helped preserve this cultural heritage.

Keys to Success:

- Inherited local wisdom and herbal practices.
- Shared preparation methods across tribes
- Scientific research for documentation and outreach

B. International Case Studies

1. Vinh Chau Village, Soc Trang – Vietnam

Flagship Products: Herbal Mangrove Salt and Mangrove Fruit Chips
Actors: Coastal Women's Cooperative

Success Story:

The cooperative transforms *Sonneratia caseolaris* and *Avicennia marina* into healthy fruit chips and mangrove-flavored herbal salt, with government-backed training and support for women's economic empowerment.

Keys to Success:

- Product innovation based on local biodiversity.
- Support from research institutions
- Market access via cooperatives and digital platforms

2. Ban Klong Khut Village, Trat – Thailand

Flagship Products: Mangrove Honey and Mud Crab Farming

Actors: Women's Groups and Fisherfolk

Success Story:

Local women and fishers harvest honey from *Apis dorsata* and farm mud crabs (*Scylla sp.*) using eco-friendly methods. These products are promoted via local festivals and exported to neighboring countries.

Keys to Success:

- Technical training and product development
- Attractive packaging and food safety certification
- Support from local government and NGOs

3. Matang Mangrove Forest, Perak – Malaysia

Flagship Products: Wild Honey, Herbal Drinks, and Natural Soaps
Actors: Fisherfolk Cooperatives

Success Story:

Managed by local cooperatives, communities harvest wild honey and use Rhizophora leaves and fruit for drinks and soaps. Products are certified halal and organic, providing jobs while preserving the forest.

Keys to Success:

- Community-based management
- Ecosystem service payment models
- Job creation and sustainability linkage

4. Barangay Parang, Calatagan – Philippines

Flagship Products: Mangrove Vinegar and Natural Dyes
Actors: Local Coastal Communities

Success Story:

Locals ferment Sonneratia fruit into vinegar and use mangrove bark for natural dyes. Supported by NGOs and universities, the products are now marketed locally and at community exhibitions.

Keys to Success:

- Training in hygiene and entrepreneurship
- Women and youth engagement
- Local marketing and education integration

VIII.

CLOSING

Mangrove forests are not just natural shields against abrasion and tsunamis—they are also vital sources of life and livelihoods for coastal communities. Through sustainable use of NTFP such as pedada fruit, nypa sap, mangrove leaves, bark, and forest honey, communities can create high-value products without harming the trees or ecosystems.

This guide has provided insights into:

1. The ecological importance of mangrove ecosystems as a foundation for sustainability
2. Various mangrove species and their NTFP potentials
3. Eco-friendly harvesting techniques that protect the trees
4. Processing techniques for food, drinks, crafts, and natural products
5. Marketing and packaging strategies to boost competitiveness.
6. Real-life community-based conservation and entrepreneurship success stories

With a foundation of sustainability, local identity, and market readiness, mangrove-based NTFP enterprises can become new economic pillars for coastal villages—ensuring the preservation of ecosystems for future generations.

This guide is not the final word—it is the first step in a broader movement:

1. To manage mangroves not just for today, but for a better tomorrow
2. To shift mindsets from exploitation to care and wise harvesting.
3. To empower coastal communities as key players in eco-friendly enterprise

We invite:

1. Coastal communities to explore local potential, form business groups, and innovate.
2. Advisors and extension workers to mentor and build community capacity.
3. Government and private sector to support policy, training, markets, and finance.
4. Youth return to their villages with pride and build businesses rooted in local wisdom.